Audrey Horn

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EDUCATION

Cornell University **Expected 2029**

Intended B.S. in Applied & Engineering Physics; GPA: 3.61/4.0

- **Relevant Coursework:** Digital Logic & Computer Organization; Computer Science; Electricity & Magnetism; Experimental Physics; Quantum Information Hardware; Differential Equations; Linear Algebra; Waves & Oscillations; Lasers & Photonics; Multivariable Calculus; Mechanics & Special Relativity.

TECHNICAL SKILLS

Hardware & Photonics: SolidWorks, ANSYS (FEA), oscilloscopes, circuit prototyping & troubleshooting, pulsed laser systems, soldering, electromechanical integration.

Digital & Embedded: Verilog, Quartus Prime, FPGA synthesis & simulation, CMOS logic, ESP32, embedded C/C++.

Programming & Analysis: Python (NumPy, SciPy, Pandas, Matplotlib, Astropy, Qiskit), Java, Git/GitHub, Linux.

RESEARCH & TECHNICAL EXPERIENCE

Cornell Hyperloop Project Team **Ithaca, New York**

Magnetics Hardware Engineer *Nov 2025 – Present*

- Designed, built, and debugged an ESP32-controlled coil-winding system integrating stepper motors, motor drivers, mechanical drive components, and embedded C/C++ control software; implemented configurable winding speed, turn count, and traversing motion while troubleshooting electromechanical integration.
- Manufacture electromagnets for the pod's electromagnetic suspension system and support linear induction motor integration and testing.
- Developed electromechanical hardware for Hyperloop systems using SolidWorks and Ansys, considering structural loads, component interfaces, and system-level constraints; contributed to the team's 5th-place finish in the Design-only division at European Hyperloop Week 2026.

Cornell University, Quantum Computing Association **Ithaca, New York**

Quantum Computing Researcher (Decoding Subteam) *Apr 2026 – Present*

- Investigate quantum error-correction decoding algorithms, including belief propagation, minimum-weight perfect matching (Blossom), and Union-Find, with emphasis on hardware constraints and physical qubit error models.
- Collaborate with theory and hardware subteams to translate decoding algorithms into FPGA-suitable architectures, considering latency, resource utilization, and integration with real-time quantum error-correction pipelines.

Cornell University, Applied & Engineering Physics **Ithaca, New York**

Laboratory Assistant (Lasers & Photonics, AEP 1100) *Aug 2026 – Present*

- Instruct students in operating, testing, and troubleshooting electronic and optical laboratory systems, enforcing equipment safety protocols and instilling experimental procedures and measurement techniques.
- Guide students through complex circuit analysis, electronic measurements, experimental troubleshooting, and laboratory calculations through discussions and office hours.

Cornell Extreme Astrophysics Group **Ithaca, New York**

Undergraduate Researcher (Nexus Scholar, Summer 2026) *May 2026 – Present*

- Curated a validated database of 126 FRB events from 200+ screened datasets and developed automated Python pipelines for data ingestion, normalization, coordinate conversion, uncertainty propagation, and source traceability.
- Recomputed extragalactic dispersion measures using the NE2025 electron-density model and produced scattering/redshift diagnostics to test theoretical predictions.

SELECTED PROJECTS

Two-Function Calculator | Verilog, Quartus Prime, FPGA

- Designed and implemented a 16-bit add/multiply calculator in gate-level Verilog, integrating custom adders, multipliers, and seven-segment display logic on an Intel Cyclone V FPGA.
- Optimized and compared circuit area and critical-path delays through synthesis and timing analysis; verified functionality using exhaustive, randomized, and X-propagation testbenches.

Nitrogen Laser Pulse Amplifier | High-Voltage Electronics, SCRs, Oscilloscope, Soldering

- Built and tested a high-voltage pulse amplifier for a pulsed nitrogen laser, using an SCR-triggered discharge circuit and 1:4 pulse transformer to generate approximately 1.6 kV trigger pulses for the laser thyratron.
- Assembled and troubleshot rectification, capacitor charging, diode, resistor, and transformer circuitry, and verified pulse behavior using a voltage-divider output and oscilloscope measurements.